

Carmen Lucca DiMario

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EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Graduation:

Master of Science in Computer Science

GPA: 4.0

May 2027

Bachelor of Science in Computer Science

GPA: 3.97

May 2026

TECHNICAL SKILLS

Programming Languages: Go, TypeScript, JavaScript, Python, Java, C, C++, PHP, SQL, Swift, Bash

Web & Full-Stack: React, Next.js, Symfony, Laravel, PostgreSQL, RESTful APIs, Auth0/JWT

Development Frameworks: Agile, Scrum

Tools and Technologies: UNIX, AWS, Git, ROS2 (Robot Operating System), STM32 Microcontrollers

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Developed a system test suite for a proprietary avionics communication interface, validating real-time engine metric transmission and ensuring protocol correctness under embedded constraints
- Implemented Clang compiler support as a replacement for legacy ARMcc toolchains
- Extended embedded library support for 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Redesigned and improved the testing framework for the entire Accuro software suite, increasing reliability
- Integrated the PETSc linear algebra library into the Accuro application, reducing complex computation time
- Analyzed the advanced C/C++ codebase to identify differences in LET (Linear Energy Transfer) calculations through various substrates

ivDash: Full Stack Development Intern

May 2024 - August 2024

- Led development of an admin dashboard consolidating functionality for doctors, nurses, and product providers across a digitalized healthcare distribution platform
- Built interactive web interfaces in Next.js against an existing Symfony PHP backend
- Streamlined Auth0 JWT authentication to work across both the Symfony PHP and Next.js applications

PROJECT EXPERIENCE

Dilithium: Post-Quantum Cryptocurrency

January 2026 - Present

- Built a peer-to-peer proof-of-work blockchain from first principles in Go (node, wallet CLI, and CPU/GPU miner) signing every transaction with CRYSTALS-Dilithium Mode3 (NIST FIPS 204, Level 3) for quantum-safe security from block zero
- Implemented competitive mining across a live P2P network with automatic block and transaction synchronization
- Exposed a REST API with UPnP port forwarding and cross-platform builds for Linux, macOS, and Windows

PFactor: iOS Application

August 2025 - Present

- Built and shipped an AI oral-exam simulator for FAA checkride prep end to end, owning product design, backend, and client
- Architected a serverless Convex backend handling authentication (Sign in with Apple), StoreKit 2 subscriptions, and graceful offline degradation via automatic local-scoring fallback
- Created an adaptive examiner engine scoring responses 1-5, generating targeted follow-ups on weak areas, and delivering per-category performance debriefs

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R. et al. (incl. DiMario, C.) "Robust Real-Time UAV Target Tracking with Onboard Vision-Based Yaw Control." Beyond: Undergraduate Research Journal, Vol. 9, 2025. <https://commons.erau.edu/beyond/vol9/iss1/6>

EXTRACURRICULAR ACTIVITIES

- **ERAU Division II Cross-Country/Track:** 7-8 practices, approximately 30 hours, weekly
- **FAA Licenses:** Part 107 Remote Pilot Certification

<https://dimario.dev>